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Zorevunersen in Children and Adolescents with Dravet Syndrome

Authors

Linda Laux, M.D.*, Joseph Sullivan, M.D.*, M Scott Perry, M.D.*, Andreas Brunklaus, M.D.*, Archana Desurkar, M.D., John M Schreiber, M.D., Colin M Roberts, M.D., Kelly G Knupp, M.D., James W Wheless, M.D., Elaine C Wirrell, M.D., Pam Ventola, Ph.D., Fei Wang, Ph.D., Meena, Ph.D., Jessie Lynch, M.S., Kimberly A Parkerson, M.D., Ph.D., Barry Ticho, M.D., Ph.D., J Helen Cross, M.B., Ch.B., Ph.D., on behalf of the MONARCH and ADMIRAL, and SWALLOWTAIL and LONGWING study groups

Affiliations

From the Ann & Robert H. Lurie Children's Hospital of Chicago, Northwestern University Feinberg School of Medicine, Chicago, IL, USA (L.L.); Benioff Children's Hospital, University of California San Francisco, Department of Neurology, San Francisco, CA, USA (J.S.); Cook Children's Medical Center, Fort Worth, TX, USA (M.S.P.); University of Glasgow, Glasgow, UK (A.B.); Sheffield Children's Hospital NHS Foundation Trust, Sheffield, UK (A.D.); Children's National Hospital, Washington D.C., USA (J.M.S.); Oregon Health & Science University, Portland, OR, USA (C.M.R.); Children's Hospital Colorado, Aurora, CO, USA (K.G.K.); Le Bonheur Children's Hospital, University of Tennessee Health Science Center, Memphis, TN, USA (J.W.W.); Mayo Clinic, Rochester, MN, USA (E.C.W.); Cogstate Ltd., New Haven, CT, USA (P.V.); Stoke Therapeutics, Bedford, MA, USA (F.W., M., J.L., K.A.P., B.T.); University College London NIHR BRC Great Ormond Street Institute of Child Health, London, UK (J.H.C.).

Corresponding author

Dr Cross can be contacted at h.cross@ucl.ac.uk or at University College London, Great Ormond Street Institute of Child Health, 30 Guilford St, London, WC1N 1EH, United Kingdom.

Notes

*Drs Laux, Sullivan, Perry, and Brunklaus contributed equally to this article.

Investigators in the MONARCH and ADMIRAL and SWALLOWTAIL and LONGWING study groups are listed in the Supplementary Appendix.

Abstract

Background

Dravet syndrome is a severe developmental and epileptic encephalopathy primarily caused by *SCN1A* haploinsufficiency. Here, we describe the effects of zorevunersen, an antisense oligonucleotide designed to upregulate Na_v1.1 sodium channels, in patients with Dravet syndrome.

Methods

Patients with Dravet syndrome aged 2–18 years on standard antiseizure medications were enrolled in two Phase 1/2a, open-label, multi-center studies (MONARCH/ADMIRAL). Patients received single (SAD) or multiple (MAD) ascending doses of zorevunersen ranging from 10–70 mg (SAD) or 20–70 mg (MAD). Eligible patients rolled over to the SWALLOWTAIL/LONGWING open-label extension (OLE) studies, where they continued receiving zorevunersen (≤45 mg) every 4 months. Safety, pharmacokinetics, and clinical effects of zorevunersen were evaluated.

Results

81 patients were enrolled across MONARCH/ADMIRAL. As of May 30, 2025, 75 patients had entered the OLE studies. Most TEAEs were mild or moderate. The most common TEAE in MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING was post lumbar puncture syndrome (24.7%) and CSF protein increase (45.3%), respectively. One participant had suspected unexpected serious adverse reactions, one discontinued the study, two died due to SUDEP, and one died of malnutrition. Patients who received 70 mg zorevunersen (SAD/MAD) in MONARCH/ADMIRAL followed by ≤45 mg in SWALLOWTAIL/LONGWING experienced median seizure reductions of 58.82%–90.91% over the first 20 months of the OLEs. Improvements were seen in overall clinical

status, quality of life, and adaptive behavior following continued dosing for up to 36 months in the OLEs.

Conclusions

These results support the continued development of zorevunersen as a potential disease-modifying treatment for Dravet syndrome. (Funding: Stoke Therapeutics; Trial registration, MONARCH:NCT04442295; ADMIRAL:ISRCTN99651026; SWALLOWTAIL:NCT04740476; LONGWING:ISRCTN12811235)

Introduction

Dravet syndrome is a severe developmental and epileptic encephalopathy caused primarily by variants in one copy of the voltage-gated sodium channel type 1 alpha subunit (*SCN1A*) gene, resulting in *SCN1A* haploinsufficiency and reduced Na_v1.1 sodium channel expression.^{1,2} These channels are highly expressed in GABAergic inhibitory interneurons and some glutamatergic neurons in the brain.^{3,4} Reductions in Na_v1.1 disrupt the excitatory-inhibitory balance, leading to general hyperexcitability and seizures.^{3,5}

The clinical phenotype of Dravet syndrome includes a spectrum of symptoms that emerge early and evolve.⁶ Most patients experience cognitive deficits, communication and behavioral impairments, motor dysfunction, growth delays, and autistic traits.⁷⁻¹³ Difficulties with feeding, poor appetite, and weight loss are also common.^{8,14} These impairments reduce quality of life (QoL) for patients and caregivers.¹⁵ Additionally, the risk of sudden unexpected death in epilepsy (SUDEP) is higher in patients with Dravet syndrome compared with the general population with epilepsy.^{16,17}

The current standard of care (SOC) for Dravet syndrome includes antiseizure medications (ASMs), dietary therapy, and/or neuromodulation with the goal of optimized seizure control.¹⁸ However, these treatments fail to control seizures in most patients, and their effect on non-seizure symptoms is minimal.^{9,19} Moreover, reductions in seizure frequency may not necessarily correspond to improvements in non-seizure outcomes, which are a major concern of caregivers.^{8,19} Consequently, disease-modifying therapies directly addressing the underlying channelopathy to improve both seizure and non-seizure symptoms are needed.^{11,15,20}

Zorevunersen (STK-001), an antisense oligonucleotide (ASO), is designed to address the channelopathy underlying Dravet syndrome. In eukaryotes, precursor messenger ribonucleic acid (mRNA) matures into mRNA via splicing, the process by which introns are removed and exons joined together.^{21,22} Non-productive (poison) exons contain a premature termination codon, which,

if included in mRNA, results in its degradation.^{21,22} Zorevunersen binds to wild-type *SCN1A* pre-mRNA to prevent non-productive exon inclusion, thereby increasing productive mRNA synthesis and Na_v1.1 protein expression.^{21,23} Restoring Na_v1.1 to physiological levels in the brain could improve overall function, including cognition and behavior, in patients with Dravet syndrome. Here, we present results from the Phase 1/2a MONARCH and ADMIRAL studies, which investigated the safety, pharmacokinetics, and effects on seizures, QoL, and adaptive behavior of zorevunersen in patients with Dravet syndrome, and interim results (May 30, 2025 data cut) from the ongoing SWALLOWTAIL and LONGWING open-label extensions (OLE) studies.

Methods

Study design

MONARCH and ADMIRAL were open-label, multi-center studies conducted in the USA and UK, respectively, to facilitate broader recruitment/access. MONARCH/ADMIRAL patients were followed for ≥ 6 months after receiving the last dose of zorevunersen. Those who completed the studies and met enrollment criteria continued receiving zorevunersen in SWALLOWTAIL and LONGWING. **Figure S1** depicts the study timeline.

MONARCH included a single ascending dose (SAD) part, in which patients received a single dose of 10, 20, 30, 45, or 70 mg zorevunersen on Day 1, and a multiple ascending dose (MAD) part, in which patients received repeating doses at 20, 30, or 45 mg, administered on Days 1, 29, and 57, to evaluate dose level and dosing interval (see the Supplementary Appendix for sentinel dosing). ADMIRAL had MAD cohorts with dose levels of 30, 45, and 70 mg, administered on Days 1, 57, and 85. To evaluate a reduced dosing schedule at the highest dose level, six patients in the MAD 70 mg cohort received only two doses of zorevunersen, on Days 1 and 57.

The OLE dosing regimen consists of three initial doses, administered on Day 1, Month 4 (Week 16), and Month 8 (Week 32). Patients who complete these initial three doses receive additional doses every 4 months (16 weeks) until study discontinuation. All OLE patients are now receiving continued doses of 45 mg zorevunersen. Details on intrathecal zorevunersen administration, dosing history, and End-of-Study/Follow-up visits are provided in the Supplementary Appendix.

Study oversight

Review boards or ethics committees at each participating institution approved the study protocols. Studies were conducted in accordance with the ethical principles set by the Declaration of Helsinki and the International Council on Harmonisation Good Clinical Practice guidelines. All patients or their legal representatives provided written informed consent.

The sponsor (Stoke Therapeutics) provided the study drug, developed the study protocols, and participated in data collection. All authors contributed to data collection, analysis, and interpretation, and affirm adherence to study protocols and data accuracy/integrity. The sponsor made the decision to publish the manuscript in agreement with all authors, who reviewed and approved the final version. Confidentiality agreements between the authors and sponsor were signed.

Patients

MONARCH enrolled patients aged 2 to 18 years (inclusive); ADMIRAL enrolled patients aged ≥ 2 to <18 years. Otherwise, eligibility criteria were comparable. Eligible patients had to experience ≥ 4 convulsive seizures during the initial 28 days of the Observation Period despite at least 1 concomitant ASM and prior use of ≥ 2 ASMs. All MONARCH/ADMIRAL patients had an established diagnosis of Dravet syndrome confirmed by an external consortium with expertise in the condition, together with a pathogenic, likely pathogenic, or variant of uncertain significance in *SCN1A*.

To enter SWALLOWTAIL or LONGWING, patients must have completed MONARCH or ADMIRAL within the prior 4 weeks (unless approved by the sponsor) with an acceptable safety profile per investigator judgement. Study protocols include detailed inclusion and exclusion criteria.

Endpoints

MONARCH/ADMIRAL primary endpoints included the safety profile, plasma pharmacokinetics (PK), and cerebrospinal fluid (CSF) exposure of single and/or multiple doses of zorevunersen. For SWALLOWTAIL/LONGWING, the primary endpoint was the safety profile of multiple doses of zorevunersen.

Across studies, common secondary endpoints included change from baseline in convulsive seizure frequency calculated over 1-month (28-day) intervals, change in overall clinical status as measured by the Clinical and Caregiver Global Impression of Change (CGI-C and CaGI-C, respectively) assessments, and change from baseline in patient QoL as measured by the EuroQoL visual analogue scale (EQ-VAS) component of the EuroQoL 5-dimension (Youth) (EQ-5D-Y) instrument. Changes in the CGI-C and CaGI-C were rated on a 7-point Likert scale ranging from 1 for “very much improved” to 7 for “very much worse”.²⁴ The EQ-VAS is a vertical scale from 0 to 100, with 0 representing the worst imaginable health to 100 representing the best imaginable health.²⁵ Secondary endpoints specific to SWALLOWTAIL/LONGWING included plasma pharmacokinetics and CSF exposure to multiple doses of zorevunersen.

Exploratory endpoints included additional measures of seizure frequency and change from baseline in adaptive behavior as measured by the Vineland Adaptive Behavior Scales, Third Edition (Vineland-3) in ADMIRAL and SWALLOWTAIL/LONGWING (see the Supplementary Appendix for details on Vineland-3 domains and subdomains). Vineland-3 raw scores range from 0 to 116, with the upper limit varying by subdomain.²⁶ Caregivers have been reported to consider improvements of 1–3 raw score points as meaningful.²⁷

Statistical analysis

No formal sample size calculations or hypothesis testing were performed for MONARCH/ADMIRAL as they were primarily dose-finding studies. See the Supplementary Appendix for definitions of the safety, pharmacokinetic and clinical evaluable sets.

Seizure frequencies were calculated per 1-month (28-day) intervals throughout the studies; intervals were aligned to begin relative to the most recent dose of zorevunersen. Baseline seizure frequencies were calculated from the month (28 days) prior to the first dose in MONARCH/ADMIRAL. For the primary analysis, seizure data intervals were excluded if seizure data were available for <50% of the days within a 1-month (28-day) interval, or, in MONARCH/ADMIRAL only, for timepoints following ASM modification (**Table S1** and results in Supplementary Appendix). Sensitivity analyses of convulsive seizure frequency were performed including all collected seizure data with no exclusion for ASM modification, with the results supporting study conclusions. All seizure analyses are descriptive, with medians and 95% confidence intervals reported. The widths of the confidence intervals were not adjusted for multiplicity and should not be used to infer treatment effects.

Mixed-effects models for repeated measures (MMRM) were used to estimate outcome trajectories over time for the CaGI-C, CGI-C, EuroQol visual analogue scale (EQ-VAS), and Vineland-3. This approach enhances interpretability, maintains consistency across time points, and adjusts for potential confounding factors inherent in longitudinal data analyses. See the statistical analysis plans for additional information on MMRMs.

Missing data were not imputed; MMRMs assume that data are missing at random and include all available data. All statistical analyses were performed using Windows SAS® Version 9.4 and JMP Pro 18.2 or higher.

Results

Study population

MONARCH was conducted from June 29, 2020, to November 16, 2023; ADMIRAL was conducted from July 29, 2021, to November 07, 2023. Of the 81 patients on a background of SOC ASMs who received ≥ 1 dose of zorevunersen (MONARCH, n=62; ADMIRAL, n=19; **Figures S2 and S3**), 80 completed the studies (MONARCH, n=61; ADMIRAL, n=19). SWALLOWTAIL began on January 20, 2021; LONGWING began on May 9, 2022. As of May 30, 2025, 58/61 (95.1%) MONARCH completers rolled over into SWALLOWTAIL, and 17/19 (89.5%) ADMIRAL completers rolled over into either LONGWING or SWALLOWTAIL (**Figures S4 and S5**). Of these 75 patients, 45/59 (76.3%) in SWALLOWTAIL and 13/16 (81.3%) in LONGWING remain enrolled.

Baseline patient characteristics between MONARCH/ADMIRAL and between SWALLOWTAIL/LONGWING were comparable (**Table 1**). Mean (standard deviation) patient ages at baseline in MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING were 9.9 (5.05) and 10.4 (5.04) years, respectively. In MONARCH/ADMIRAL, 81.5% of patients were taking ≥ 3 ASMs at baseline; the most common were clobazam (70.4%), fenfluramine (49.4%), cannabidiol (44.4%), and valproate compounds (44.4%). Patient demographics were representative of the expected gender distribution and prevalence of Dravet syndrome in the respective study regions (**Table S2**).

Safety

78/81 patients (96.3%) in MONARCH/ADMIRAL and 75/75 patients (100.0%) in SWALLOWTAIL/LONGWING experienced ≥ 1 treatment-emergent adverse event (TEAE) (**Table 2**);

most TEAEs were mild or moderate in severity. The most common TEAE in MONARCH/ADMIRAL was post lumbar puncture syndrome (24.7%), while the most common TEAE in SWALLOWTAIL/LONGWING was CSF protein increase (45.3%) (**Tables S3** and **S4**). Serious TEAEs were reported in 18 patients (22.2%) in MONARCH/ADMIRAL and in 22 patients (29.3%) in SWALLOWTAIL/LONGWING (see the Supplementary Appendix) and were considered treatment-related in one patient. The most common serious TEAEs across both Phase 1/2a and OLE studies were seizure (MONARCH/ADMIRAL: 4.9%; SWALLOWTAIL/LONGWING: 5.3%) and status epilepticus (MONARCH/ADMIRAL: 1.2%; SWALLOWTAIL/LONGWING: 5.3%; **Tables S5** and **S6**).

Treatment-related TEAEs were reported in 24/81 (29.6%) and 40/75 (53.3%) patients in MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING, respectively. Treatment-related TEAEs were mild or moderate in severity, except for events in one patient in the 70 mg MAD cohort of ADMIRAL who had suspected unexpected serious adverse reactions. This patient experienced procedural vomiting and reduced fluid intake that led to hospitalization 1 day after their second dose of zorevunersen and resolved after approximately 1 week. CSF protein elevations were not observed. Approximately 2 months after receiving a third dose of zorevunersen, this patient developed cortical visual impairment, eye movement disorder, and neurological decompensation. These events were present at the end of the study and are considered treatment-related due to lack of an alternative explanation. The most common treatment-related TEAE across both Phase 1/2a and OLE studies was CSF protein increase (MONARCH/ADMIRAL: 13.6%; SWALLOWTAIL/LONGWING: 44.0%; Supplementary Appendix, **Tables S7–S10**). No patients experienced TEAEs of hydrocephalus or increased intracranial pressure; one patient experienced a TEAE of CSF pleocytosis with a CSF white blood cell count of 11 μ L. The Supplementary Appendix and **Tables S11** and **S12** summarize TEAEs related to CSF collection or study drug administration.

One SWALLOWTAIL/LONGWING patient experienced a TEAE that led to study drug discontinuation/study withdrawal (see the Supplementary Appendix).

Two patients (one in MONARCH and one in SWALLOWTAIL) died due to SUDEP, consistent with their underlying disease, and one SWALLOWTAIL patient died of malnutrition (see the Supplementary Appendix).

Pharmacokinetics

In MONARCH/ADMIRAL, plasma zorevunersen levels generally increased proportionally with dose (**Figure S6**). Zorevunersen was detected in CSF up to 6 months after the last dose in all cohorts. In the MAD cohorts, zorevunersen levels in CSF increased 3 months after the last dose, suggesting zorevunersen accumulation in CSF following monthly/bimonthly dosing (**Figure S7**).

In SWALLOWTAIL/LONGWING, following zorevunersen administration every 4 months, no apparent plasma accumulation was observed across all dose levels. Conversely, mean trough CSF concentrations generally increased with subsequent dosing over the first year before reaching close to steady state levels.

See the Supplementary Appendix, **Figures S6 and S7**, and **Table S13** for additional PK data.

Seizure frequency

Seizure data were available for most patients at all timepoints (**Table S1** and results in Supplementary Appendix). The median (range) Phase 1/2a baseline convulsive seizure frequency per 1-month (28 days) was 17.0 (4.0–2,335.4) across patients (n=77) and ranged from 8.3–64.0 across dose cohorts (**Tables 1 and S14**). **Figures 1A and 1B** illustrate the median change from

baseline in convulsive seizure frequencies over 1-month (28-day) intervals across different MONARCH/ADMIRAL dose cohorts. Median reductions in convulsive seizure frequency at 3 and 6 months after the last dose were 42.81% and 57.30%, respectively, in patients who received a single dose of 70 mg zorevunersen, and 84.80% and 73.61%, respectively, in patients who received 2 or 3 doses of 70 mg zorevunersen (Supplementary Appendix, **Figures S8 and S9**). The proportions of patients experiencing $\geq 50\%$, $\geq 75\%$, and 100% reduction in convulsive seizure frequency, convulsive seizure-free days, and total seizure frequencies are described in the Supplementary Appendix and **Tables S15–S17**.

Percent change in convulsive seizure frequency from Phase 1/2a baseline among SWALLOWTAIL/LONGWING patients are presented in **Figures 1C and 1D**. In patients who received doses of < 70 mg zorevunersen in Phase 1/2a, median reductions from Phase 1/2a baseline ranged from 6.94%–77.38% over 1-month (28-day) intervals through Month 36 of the OLEs (**Figure 1C**). Median reductions in patients who received single or multiple doses of 70 mg zorevunersen in MONARCH/ADMIRAL ranged from 58.82%–90.91% over 1-month (28-day) intervals through Month 20 of the OLEs (**Figure 1D**).

Overall clinical status and QoL

In MONARCH/ADMIRAL, clinicians and caregivers rated 84.2% and 94.8% of patients who received single or multiple doses of 70 mg zorevunersen, respectively, as “minimally improved” to “very much improved” at 6 months after last dose (**Figures 2A and 2B**).

In SWALLOWTAIL/LONGWING, clinicians and caregivers rated 94.7% of all enrolled patients as “minimally improved” to “very much improved” at Month 36 compared with OLE baseline (**Figure 2C**; Supplementary Appendix, **Figures S10 and S11**).

Exploratory cross-study comparisons adjusted for baseline prognostic covariates suggest that improvements in overall clinical status among patients with Dravet syndrome treated with zorevunersen appear to differ from the minimal change detected among patients from the BUTTERFLY natural history study (**Figure S12**). The QoL data are consistent with improved outcomes (Supplementary Appendix, **Figures S13 and S14**).

Adaptive behavior

Figures 3A and 3B display changes in Vineland-3 subdomains among ADMIRAL patients who rolled over into LONGWING and among OLE patients, respectively.

Improvements in Expressive and Receptive Communication subdomains, measured as least squares mean change in raw scores from ADMIRAL baseline, were 3.29 and 5.21, respectively, at the end of ADMIRAL (Month 9) and 6.55 and 10.00, respectively, by Month 17 of LONGWING (**Figure 3A**). At Month 36 of the OLEs, the least squares mean change in raw scores from OLE baseline in Expressive and Receptive Communication were 7.56 and 6.06, respectively (**Figure 3B**).

Exploratory cross-study comparisons adjusted for baseline prognostic covariates suggest that changes in adaptive behavior among patients with Dravet syndrome treated with zorevunersen appear to differ with respect to the minimal improvements in BUTTERFLY patients; BUTTERFLY is a prospective study designed to capture natural history of the disease (**Figure S15**).²⁸

Real-life impacts on activities of daily living associated with the clinical data described here are conveyed by the clinical improvements recorded in an ADMIRAL/LONGWING patient (see the Supplementary Appendix for details). See Supplementary video of this patient before/after zorevunersen treatment.

Discussion

Treatment of patients with zorevunersen did not result in any apparent safety signal. The results reported here support the continued development of zorevunersen as a potential disease-modifying treatment for Dravet syndrome. Most treatment-related TEAEs were mild or moderate, with the most common across MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING being elevated CSF protein as observed on routine laboratory screening. Similar CSF protein elevations have been reported in other FDA-approved intrathecal ASOs.²⁹⁻³¹ Increased CSF protein following repeated nusinersen dosing has been reported and may be a possible therapeutic marker.³² No patients experienced TEAEs of hydrocephalus or increased intracranial pressure. One patient experienced CSF pleocytosis. Ongoing safety monitoring includes neurological examinations and CSF laboratory testing.

Zorevunersen treatment was associated with improvements in seizure outcomes, overall clinical status, QoL, and adaptive behavior in patients with Dravet syndrome refractory to SOC treatment with ASMs. While the study design did not support formal statistical comparisons among dose groups, median seizure reductions were largest in patients receiving multiple initial doses of 70 mg, consistent with a dose-effect relationship. Continued treatment in SWALLOWTAIL/LONGWING appeared to show an association with stabilized reductions in convulsive seizure frequency over 36 months. Overall, the seizure data suggest differences with respect to the natural history study,²⁸ in which patients with Dravet syndrome on best-available SOC treatment on average showed no improvement in convulsive seizure frequency through 24 months.

Consistent with zorevunersen's mechanism of upregulating *SCN1A* in the context of the most common underlying cause of Dravet syndrome, treatment with zorevunersen appeared to be associated with improvements in overall clinical status, QoL, and adaptive behavior. No approved

therapies address the underlying cause of Dravet syndrome, and caregivers have indicated that alleviating non-seizure aspects are important for new treatments.³³ Continuing improvements were noted across Vineland-3 subdomains. These include Expressive and Receptive Communication, which caregivers view as one of the most important disease aspects to change following treatment.^{27,33} The changes in adaptive behavior and overall clinical status among patients with Dravet syndrome treated with zorevunersen appeared to be different from those observed in the natural history study and warrant further evaluation as part of ongoing clinical development.

Patient demographics in MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING reflected the expected gender distribution and prevalence of Dravet syndrome in the respective study regions (**Table S2**). Study limitations include the open-label design, lack of placebo or sham control, small sample sizes, lack of formal power calculations, uncorrected multiple comparisons, and variability in baseline measures (e.g., seizure frequency across dose groups), all of which prevent us from drawing conclusions about efficacy. Moreover, Vineland-3 and QoL measures may be susceptible to caregiver bias due to the open-label design. As MONARCH/ADMIRAL and SWALLOWTAIL/LONGWING were primarily safety and tolerability studies, information on clinical outcomes requires further evaluation in a controlled and adequately powered study. Another limitation is generalizability: restricting eligibility to patients with a minimum number of convulsive seizures may limit applicability to those with fewer seizures.

Real-world implementation of zorevunersen treatment may involve considerations such as sedation for intrathecal injections, interdisciplinary care needs, and ongoing monitoring to optimize patient outcomes.

Overall, these data support the potential for disease modification with zorevunersen. Further assessment in a randomized, controlled Phase 3 trial is underway. These data also support

the selection of initial and continued doses of 70 mg and 45 mg zorevunersen, respectively, in the ongoing Phase 3 trial.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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Figure Legends

Figure 1. Median percent change from Phase 1/2a baseline in convulsive seizure frequency in MONARCH and ADMIRAL and SWALLOWTAIL and LONGWING

A. Median percent change from baseline in convulsive seizure frequency in MONARCH SAD cohorts

B. Median percent change from baseline in convulsive seizure frequency in MONARCH and ADMIRAL MAD cohorts

C. Median percent change from Phase 1/2a baseline in convulsive seizure frequency in SWALLOWTAIL and LONGWING[†]

D. Median percent change from Phase 1/2a baseline in convulsive seizure frequency in SWALLOWTAIL and LONGWING in patients who received 1, 2, or 3 doses of 70 mg zorevunersen in MONARCH and ADMIRAL

Ph1/2a data cut: December 12, 2023 (after End of Study). OLE data cut: May 30, 2025.

Panels A and B show the median percent change from baseline in convulsive seizure frequency in MONARCH SAD cohorts and combined MONARCH and ADMIRAL MAD cohorts, respectively. Panel C shows the median percent change from Ph1/2a baseline in convulsive seizure frequency up to Month 36 of the OLEs in all enrolled patients who received doses of <70 mg zorevunersen and those who received 70 mg (1, 2, or 3 doses) zorevunersen in MONARCH and ADMIRAL. Panel D compares median reductions in convulsive seizure frequency up to Month 20 of the OLEs in patients who received 1, 2, or 3 doses of 70 mg zorevunersen in MONARCH and ADMIRAL.

Convulsive seizures were defined as hemiclonic, focal with motor signs, focal to bilateral tonic-clonic, generalized tonic-clonic, tonic, tonic/atonic (drop attacks), or clonic types. Seizure frequencies were calculated for 1-month (28-day) intervals in the studies.

Error bars represent 95% CIs. The widths of the CIs have not been adjusted for multiplicity and should not be used to infer treatment effects.

*MONARCH MAD cohorts were dosed on Days 1, 29, and 57. ADMIRAL included 30, 45, and 3-dose 70 mg MAD cohorts with drug administration on Days 1, 57, and 85 and a 2-dose 70 mg MAD cohort with drug administration on Days 1 and 57.

[†]70 mg 1-dose patients received 30 or 45 mg doses of zorevunersen at Months 1, 5, 9, 13, and 17 of the OLE, while 70 mg 2- or 3- dose patients received 45 mg doses of zorevunersen at Months 1, 5, 9, 13, and 17 of the OLE. Patients were not included in 6 month (6M) time point after the last Ph1/2a dose if they did not enter OLEs. No exclusions were made for ASM modification in the OLEs.

ASM, antiseizure medication; BL, baseline; CI, confidence interval; M, month; MAD, multiple ascending dose; OLE, open-label extension; Ph1/2a, Phase 1/2a; SAD, single ascending dose.

Figure 2. Overall clinical status in MONARCH and ADMIRAL and in SWALLOWTAIL and LONGWING

A. Change in overall clinical status at the end of MONARCH and ADMIRAL (6 months after last dose) in patients receiving doses of <70 mg zorevunersen

B. Change in overall clinical status at the end of MONARCH and ADMIRAL (6 months after last dose) in patients receiving 1, 2, and 3 doses of 70 mg zorevunersen

C. Change in overall clinical status at Month 36 across patients enrolled in SWALLOWTAIL and LONGWING

Ph1/2a data cut: December 12, 2023 (after End of Study). OLE data cut: May 30, 2025.

CGI-C and CaGI-C assessments were scored on a Likert scale with seven response options: 1 (Very much improved), 2 (Much improved), 3 (Minimally improved), 4 (No change), 5 (Minimally worse), 6 (Much worse), or 7 (Very much worse).

Error bars represent 95% CIs. The widths of the CIs have not been adjusted for multiplicity and should not be used to infer treatment effects. Comparison of dose groups was post hoc.

CaGI-C, Caregiver Global Impression of Change; CGI-C, Clinical Global Impression of Change; CI, confidence interval; OLE, open-label extension.

Figure 3. Change in Vineland-3 subdomain raw scores in ADMIRAL through LONGWING and in SWALLOWTAIL and LONGWING

A. LS mean change in Vineland-3 subdomain raw scores of patients (all dose cohorts) enrolled in ADMIRAL through LONGWING*

B. LS mean change in Vineland-3 subdomain raw scores of patients enrolled in SWALLOWTAIL and LONGWING†

Ph1/2a data cut: December 12, 2023 (after End of Study). OLE data cut: May 30, 2025.

The analyses focused on the seven subdomains without minimum age criteria and the one subdomain with minimum age criterion of 2 years. For the remaining three subdomains, baseline age-equivalent scores in ADMIRAL were at or below floor in 9/17 patients (52.9%) in Written, 15/17 patients (88.2%) in Domestic, and 13/17 patients (76.5%) in Community, and in the OLEs were at or below floor in the OLEs in 26/71 patients (36.6%) in Written; 49/71 patients (69.0%) in Domestic; and 48/71 patients (67.6%) in Community.

Error bars represent 95% CIs. The widths of the CIs have not been adjusted for multiplicity and should not be used to infer treatment effects.

*Mixed-effects models for repeated measures were employed to analyze data from ADMIRAL (all dose cohorts) through Week 96 of LONGWING (which corresponds to Week 132 from ADMIRAL baseline). Changes from baseline in raw scores were estimated over time adjusting baseline prognostic covariates. ADMIRAL sample size: n=17–18 at screen.

†Mixed-effects models for repeated measures were constructed using available data from enrolled patients in SWALLOWTAIL and LONGWING. Changes from OLE baseline in raw scores were

estimated over time, adjusting baseline prognostic covariates. Combined SWALLOWTAIL and LONGWING sample sizes were n=74 at OLE baseline.

CI, confidence interval; LS, least squares; OLE, open-label extension; Vineland-3, Vineland Adaptive Behavior Scales, Third Edition.

Table 1. Baseline characteristics of study participants in MONARCH and ADMIRAL and SWALLOWTAIL and LONGWING

	MONARCH and ADMIRAL (N=81)	SWALLOWTAIL and LONGWING (N=75)
Age at screening, years		
Mean (SD)	9.9 (5.05)	10.4 (5.04)
Median (range)	10.0 (2–18)	11.0 (2–19)
Age group, n (%)		
2–12 years	46 (56.8)	42 (56.0)
≥13 years	35 (43.2)	33 (44.0)
Sex, n (%)		
Male	41 (50.6)	38 (50.7)
Female	40 (49.4)	37 (49.3)
Race,* n (%)		
Asian	5 (6.2)	5 (6.7)
Black or African American	5 (6.2)	5 (6.7)

White	71 (87.7)	66 (88.0)
Prefer not to answer	4 (4.9)	3 (4.0)
Ethnicity, n (%)		
Hispanic/Latino	10 (12.3)	10 (13.3)
Not Hispanic/Latino	70 (86.4)	64 (85.3)
Prefer not to answer	1 (1.2)	1 (1.3)
SCN1A gene variant type, n (%)		
Missense	37 (45.7)	32 (42.7)
Nonsense	44 (54.3)	43 (57.3)
Number of concomitant ASMs at Phase 1/2a baseline, n (%)[†]		
≥3	66 (81.5)	NA
≥4	41 (50.6)	NA
Concomitant fenfluramine at baseline, n (%)		
Yes	40 (49.4)	40 (53.3)
Phase 1/2a baseline convulsive seizure frequency per 28 days[‡]		
Median (range)	17 (4.0–2335.4)	NA

*Multiple selections for race could be entered.

[†]The most common ASMs in MONARCH and ADMIRAL were clobazam (70.4%), fenfluramine (49.4%), cannabidiol (44.4%), and valproate compounds (44.4%).

[‡]Includes only patients that met the clinically evaluable study criteria for seizure analysis per statistical analysis plan. n=58, n=19, and N=77 for MONARCH, ADMIRAL, and all Phase 1/2a, respectively.

ASM, antiseizure medication; NA, not applicable; *SCN1A*, voltage-gated sodium channel type 1 alpha subunit; SD, standard deviation.

Table 2. Summary of overall AEs in MONARCH and ADMIRAL and SWALLOWTAIL and LONGWING

n (%)	MONARCH and ADMIRAL	SWALLOWTAIL and LONGWING
	(N=81)	(N=75)
Any TEAE*	78 (96.3)	75 (100.0)
Treatment-related TEAE	24 (29.6)	40 (53.3)
TEAE related to CSF collection or study drug administration	43 (53.1)	45 (60.0)
Grade ≥3 TEAE	13 (16.0)	12 (16.0)
Grade ≥3 TEAE related to study drug	1 (1.2)	0 (0)
Serious TEAE	18 (22.2)	22 (29.3)
Serious TEAE related to study drug	1 (1.2)	0 (0.0)
Potential dose-limiting toxicity[†]	1 (1.2)	0 (0.0)
TEAE leading to study drug discontinuation	0 (0.0)	1 (1.3)
TEAE leading to study withdrawal	0 (0.0)	1 (1.3)
TEAE leading to death[†]	1 (1.2)	2 (2.7)

*TEAE is an AE first identified, or is identified to worsen in intensity, at a time occurring after the first dose of study drug.

†Potential dose-limiting toxicity and TEAE leading to death occurred in separate patients.

AE, adverse event; CSF, cerebrospinal fluid; OLE, open-label extension; SUDEP, sudden unexpected death in epilepsy; TEAE, treatment-emergent adverse event.